

Exploratory Data Analysis Report

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Course: Data Science

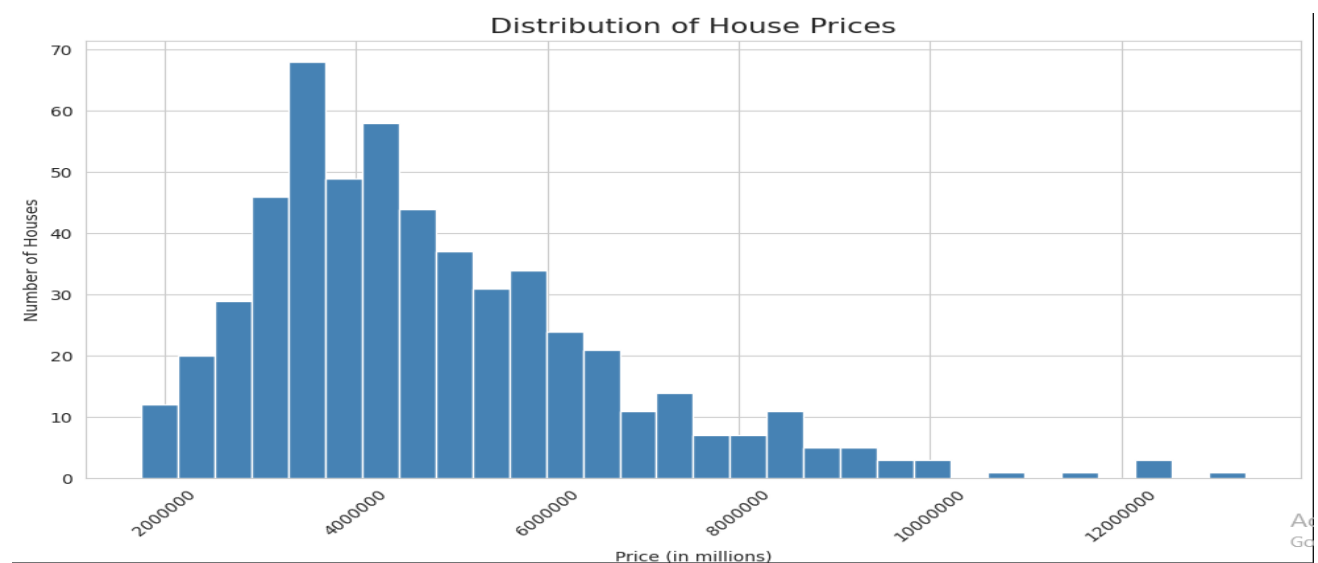
Institution: AltSchool Africa

Dataset: Housing.csv

Introduction

This report presents the findings of an Exploratory Data Analysis conducted on a real-world housing dataset containing 545 property records across 13 features. The dataset includes both numerical variables such as price, area, bedrooms, bathrooms and stories, as well as categorical variables such as main road, air-conditioning, furnishing status and preferred area. The objective of this analysis was to identify the key factors that influence house prices, uncover patterns in the data and provide practical recommendations for a hypothetical house buyer seeking value for money.

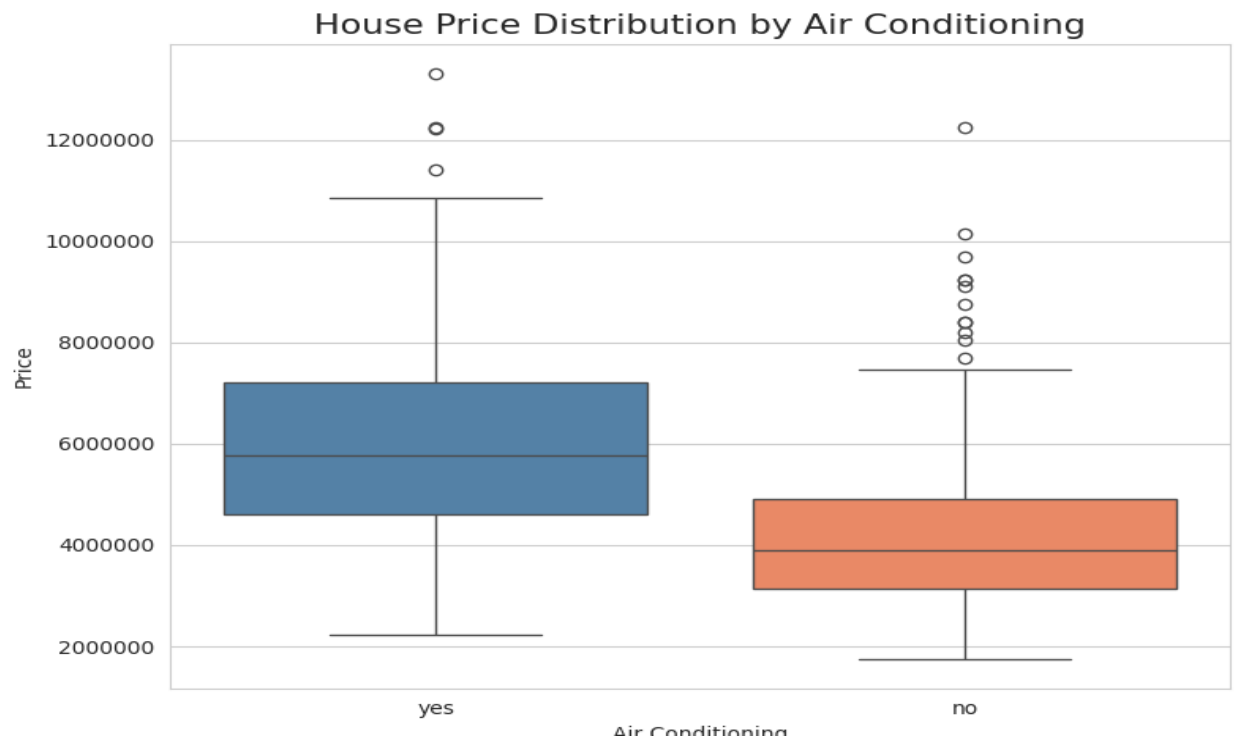
Key Finding 1 — Price Distribution



The average house price in this dataset is 4,766,729 while the median price is 4,340,000. The fact that the average is higher than the median indicates a right-skewed price distribution — clearly visible in the histogram above. This skew is caused by a small number of luxury properties priced as high as 13,300,000 which pull the average upward. As a result the median is a more reliable indicator of what a typical house costs in this market. Using the IQR method, 15 price outliers were identified above the upper fence of

9,205,000 — these represent genuine luxury properties rather than data errors and were retained for analysis.

Key Finding 2 — Key Features That Drive Price



The analysis revealed that several property features significantly impact house prices. Air conditioning was found to be the single biggest price driver — houses with AC average 6,013,221 compared to 4,191,940 without, representing a premium of 1,821,281 or approximately 43%. This difference is clearly illustrated in the box plot above where the AC box sits considerably higher than the non-AC box. Main road location adds a further premium of 1,592,873 — approximately 47% more than off-road properties. Houses in preferred areas command 1,453,747 more on average — a 33% premium. Furnishing status also plays a significant role with fully furnished homes averaging 5,495,696 compared to 4,013,831 for unfurnished properties — a difference of 1,481,865.

Key Finding 3 — Characteristics of the Most Expensive Properties

An analysis of the top 5 most expensive houses — ranging from 11,410,000 to 13,300,000 — revealed striking common features. All five properties are located on the main road, four out of five have air conditioning and four out of five are fully furnished. Most have four bedrooms and multiple stories. This confirms that the combination of prime location, cooling amenities and full furnishing creates the highest-value properties in this dataset. The correlation analysis further supports this — area and price show a

moderate positive correlation of 0.54, while the number of bathrooms and stories also show meaningful positive relationships with price.

Key Finding 4 — Bedroom and Story Distribution

The bedroom distribution analysis revealed that 3-bedroom houses are by far the most common property type — accounting for 300 out of 545 houses or exactly 55% of the dataset. This confirms that the market is dominated by medium-sized family homes. A clear positive trend was also observed between the number of stories and average price — single storey homes average 4,170,659 while four-storey properties average 7,208,450 — nearly double the price. Additionally only 66 properties have both a basement and air conditioning, yet these premium homes command an average price of 6,081,197 — well above the dataset average — demonstrating that combined amenities create disproportionate value.

Conclusion and Recommendations

This analysis of 545 housing properties has revealed that price in this market is driven not by size alone but by a combination of location, amenities and furnishing status. The distribution is right-skewed with most properties priced between 2,000,000 and 6,000,000 and a small number of luxury outliers above 9,000,000. The following recommendations are offered for a house buyer seeking the best value:

Recommendation 1 — Avoid Air Conditioned Properties and Install Independently

Air conditioning adds the largest price premium of any single feature — nearly 1,821,281 on average. A value-seeking buyer should consider purchasing a property without AC and installing it independently at a fraction of this cost — potentially saving over one million in purchase price.

Recommendation 2 — Consider Properties Away From the Main Road

Main road properties command a 47% price premium. Properties located just off the main road offer significantly lower prices while often maintaining similar access, amenities and neighbourhood quality. This represents one of the most impactful ways to reduce purchase cost.

Recommendation 3 — Purchase Unfurnished and Furnish Independently

Fully furnished homes cost on average 1,481,865 more than unfurnished equivalents. Buyers who purchase unfurnished properties and furnish them independently can achieve substantial savings while also personalising their living space to their own taste and budget.

In summary, the data strongly suggests that a buyer seeking value should prioritise properties that are off the main road, outside the preferred area, without air conditioning and unfurnished — then invest selectively in improvements after purchase to maximise both comfort and long-term property value.